POLYMER CHEMISTRY IN ACTIVEWEAR

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DESCRIPTION

A specialized field involving the production of textiles used in clothing.

Focuses on the modification of clothing surfaces in polymer chemistry.

Involves manipulation of the composition and properties of the polymer or monomers.

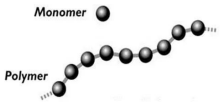
Chemists in the field of material science work with natural and synthetic fibers to change chemical properties of these surfaces.

SCIENCE

Study how complex molecules can be combined and manipulated on the molecular scale to change the properties of specific polymers and monomers.

Produced through the chemical process called polymerization which consists of chemical bonding between monomers.

Initiated through catalysts like heat, pressure, and chemical alterations on the molecular level.



Example:
High performance "dri-fit" fabric made of microfiber and polyester microfiber

Decreases the amount of sweat adhesion from the body

EFFECT ON ENGINEERING

Polymer chemistry created entirely new concentration within engineering - focuses on the design and optimization of new material.

Incorporates principles of weight, conductivity, efficiency and effectiveness against intended elemental forces (wind, rain, etc.) into clothing and other applications.

Reconciled the world of material and chemical engineering with that of fashion and design, bringing in scientific innovation in polymer technology and future nanotechnology to clothing.

HISTORY

1907 - The first synthetic polymer was invented by Leo Baekeland in a phenol-formaldehyde resin.

1940 - In WWII, there was a strong commercial polymer industry, limiting the supply of natural materials like silk and increased the production of synthetic substitutes like nylon and synthetic rubber.

2000s - Activewear became more form-fitting along with rise of body image and self care.

2010s - Technology has pushed boundaries to incorporate bodily functions into tech clothing like Outlast by Vexxed Generation, which monitors body temperature.

FUTURE APPLICATIONS

biotech clothing to regulate body heat

seamless and flat knit apparel for high-intensity sports like running

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super-absorbent diapers

healing damaged tissues

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